

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	Shaik Nuha	Reg. No.:	192472242
Section / Batch:	B.E-CSE-AI	Date:	

Code of Conduct

I Shaik Nuha(192472242) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk.Nuha

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. Source Sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Coreference Resolution Using Constraints

Given Paragraph

“John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home.”

1(a) Identify all referring expressions (mentions) and possible antecedents

A referring expression/mention is a word or phrase that refers to an entity already introduced or mentioned in the discourse.

Mention and antecedent table

Mention	Type	Possible antecedents	Final antecedent
John	Proper noun	—	John
Mary	Proper noun	—	Mary
the park	Definite noun phrase	—	Park
He	Pronoun	John, Mary	John
a ball	Noun phrase	New entity	Ball
She	Pronoun	John, Mary	Mary
it	Pronoun	John, Mary, ball	ball
The dog	Noun phrase	New entity	Dog
him	Pronoun	John, Mary, ball, dog	John
they	Plural pronoun	John + Mary, John + Mary + dog	John + Mary + dog
all	Quantifier/referring expression	John, Mary, dog	John + Mary + dog

Initial entity set

After the first sentence, the discourse contains:

$$E = \{\text{John, Mary, Park}\}$$

After the second sentence:

$$E = \{\text{John, Mary, Park, Ball}\}$$

After the fourth sentence:

$$E = \{\text{John, Mary, Park, Ball, Dog}\}$$

1(b) Apply the Coreference Constraints

The main constraints are:

1. Gender and number agreement
2. Semantic compatibility
3. Coherence

Reference 1: “He”

Sentence:

“John and Mary went to the park. He brought a ball.”

Possible antecedents:

- John
- Mary

Gender agreement

“He” is masculine.

Candidate	Gender	Result
John	Male	✓
Mary	Female	✗

Therefore:

He → John

Gender agreement eliminates Mary.

Final:

He=John

Reference 2: “She”

Sentence:

“She wanted to play with it.”

Possible antecedents:

- John
- Mary

“She” is feminine.

Candidate	Gender	Result
John	Male	✗
Mary	Female	✓

Therefore:

She=Mary

Reference 3: “it”

Sentence:

“She wanted to play with **it**.”

Possible antecedents:

- John
- Mary
- ball

“It” normally refers to an inanimate object or non-human entity in this context.

Candidate	Semantic compatibility
John	X
Mary	X
Ball	✓

The verb phrase:

“play with it”

is highly compatible with a ball.

Therefore:

it=ball

Reference 4: “him”

Sentence:

“The dog chased him excitedly.”

Possible antecedents include:

- John
- Mary
- ball
- dog

The pronoun him requires a masculine animate antecedent.

Gender and semantic compatibility

Candidate	Gender	Animate?	Suitable?
John	Male	Yes	✓
Mary	Female	Yes	X
Ball	Neutral/inanimate	No	X
Dog	Ambiguous/animal	Yes	X/unlikely

John is the strongest candidate.

Therefore:

him=John

Why not “dog”?

“The dog chased him” normally describes the dog as the agent and “him” as the person being chased.

Thus:

Dog →Chaser

John →Chased person

This provides strong semantic and syntactic compatibility.

Reference 5: “they”

Sentence:

“Finally, they all went home.”

“They” is plural.

Possible antecedent groups:

- John + Mary
- John + Mary + dog

Because the sentence says:

“they all went home”

and the paragraph explicitly introduces John, Mary and the dog as participants, the most coherent interpretation is:

they={John,Mary,Dog}

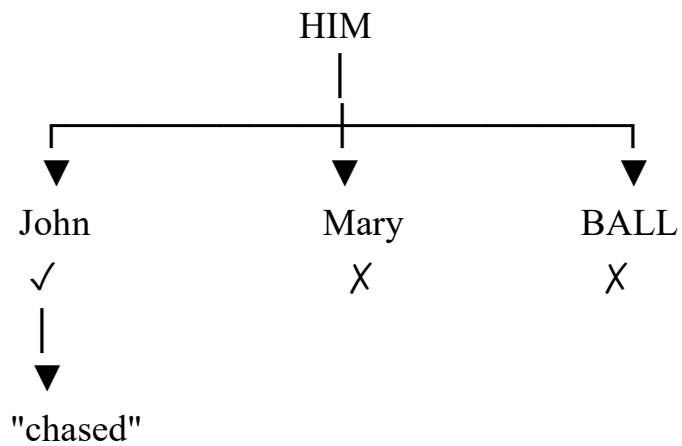
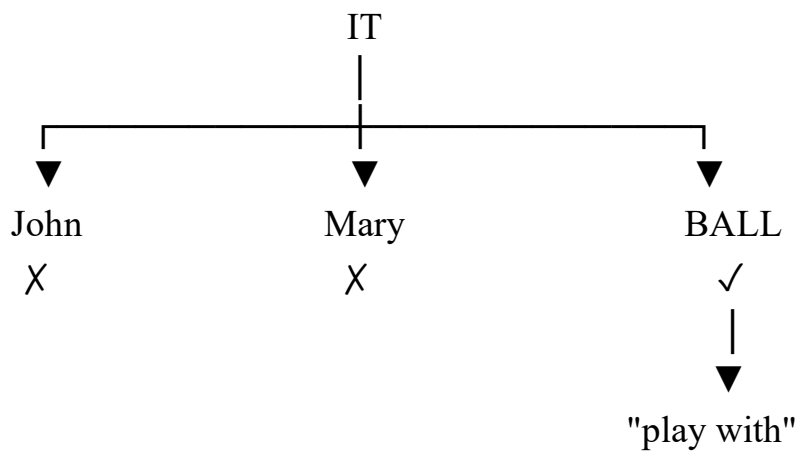
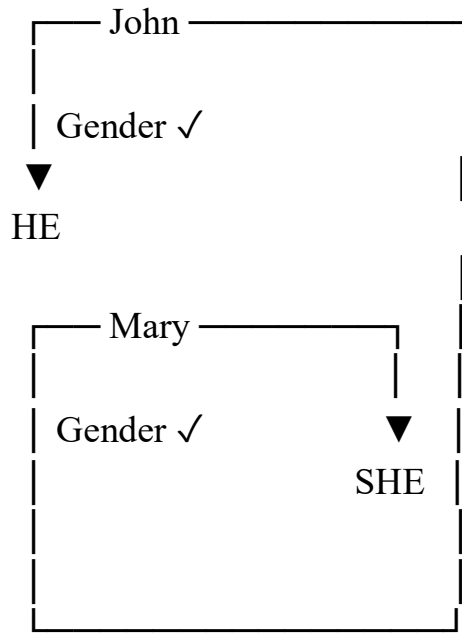
1(c) Constraint Graph / Table

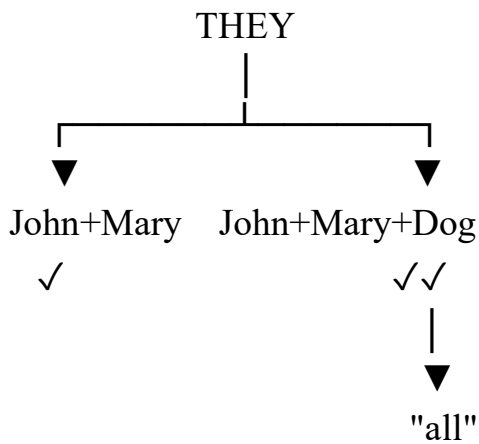
A constraint table can show how possible links are eliminated.

Coreference Constraint Table

Mention	Candidate	Gender/Number	Recency	Semantic Compatibility	Coherence	Decision
He	John	✓	✓	✓	✓	Select
	Mary	✗	✓	✓	✗	Eliminate
She	John	✗	✓	✓	✗	Eliminate
	Mary	✓	✓	✓	✓	Select
it	John	✗	—	✗	✗	Eliminate
	Mary	✗	—	✗	✗	Eliminate
	Ball	✓	✓	✓	✓	Select
him	John	✓	Moderate	✓	✓	Select
	Mary	✗	✓	✓	✗	Eliminate
	Ball	✗	✓	✗	✗	Eliminate
	Dog	Weak	✓	✗	✗	Eliminate
they	John + Mary	✓	✓	✓	✓	Possible
	John + Mary + Dog	✓	✓	✓	✓✓	Select

Simple Constraint Graph





1(d) Final Coreference Chains

The resolved chains are:

Chain 1

John → He → him → they

Chain 2

Mary → She → they

Chain 3

Ball → it

Chain 4

Dog → they

Chain 5

Park

Rewritten Paragraph with Clear References

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home. This version removes the ambiguity caused by pronouns while preserving the original meaning.

1(e) Priority Order of Constraints

A suitable priority order is:

Gender/Number > Semantic Compatibility Coherence > Recency

1. Gender and Number Agreement

This is a strong grammatical constraint.

For example:

“He” cannot normally refer to Mary.

Therefore, it should be applied early.

2. Semantic Compatibility

The antecedent should make sense with the action.

For example:

“Mary wanted to play with it.”

The ball is a natural object to play with.

John and Mary are not semantically suitable antecedents for “it” in this context.

3. Coherence

The interpretation should make the entire paragraph logically consistent.

For example:

“The dog chased him.”

John being chased by the dog is a coherent interpretation.

4. Recency

Recency is useful as a tie-breaking strategy.

It prefers a recently mentioned entity.

However, recency should not override stronger constraints.

What Happens If Recency Is Relaxed?

If recency is relaxed, the system can still correctly resolve references using:

- Gender
- Number
- Semantic compatibility
- Syntactic role
- Discourse coherence

For example, “He” still refers to John because Mary does not satisfy gender agreement.

Similarly:

“it” → ball

because semantic compatibility strongly Favors ball.

Therefore, relaxing recency may increase the number of possible candidates, but it does not necessarily produce incorrect results if the other constraints are strong.

Conclusion

Recency should generally be treated as a soft constraint, whereas grammatical agreement and semantic compatibility should often be treated as stronger constraints.

2. Constraint-Based Dialogue Generation

Conversation History

User:

“I have an important exam tomorrow but I’m not able to concentrate.”

Required dialog act:

Advise + Encourage

2(a) Discourse Planning Steps

Before generating the response, the system should perform the following steps.

Step 1: Identify the user's situation

The user has:

- An important exam
- An exam tomorrow
- Difficulty concentrating

Therefore, the response should address concentration and exam preparation.

Step 2: Identify the required dialog act

The response must perform two functions:

Advice + Encouragement

It should give a practical recommendation and motivate the student.

Step 3: Maintain entity coherence

The response should continue using:

- exam
- concentrate
- you

This keeps the conversation connected.

Step 4: Establish discourse relation

A suitable relation is Cause-Effect:

If you take a short break, you can improve your focus.

Another possibility is Elaboration:

Take a short break, then return to studying with a focused plan.

Step 5: Include required keywords

At least two of:

- focus
- break
- confident

must appear.

Step 6: Ensure length

The answer must contain:

2-3 sentences

2(b) Three Possible Responses

Response 1

Take a short break, then return to your exam preparation with one small topic at a time so you can improve your focus. You still have time to prepare, so stay confident and avoid worrying about everything at once.

Keywords:

- break
- focus
- confident

Response 2

Since your exam is tomorrow, take a short break and then study in focused 25-minute sessions to improve your concentration. You can make good progress tonight, so stay confident and focus on the most important topics first.

Keywords:

- break
- focus
- confident

Response 3

Take a short break before studying again because it can help you regain your focus for the exam. You have enough time to revise the key topics, so stay confident and concentrate on one task at a time.

Keywords:

- break
- focus
- confident

2(c) Evaluation and Best Response

Constraint	Response 1	Response 2	Response 3
Advice	✓	✓	✓
Encourage	✓	✓	✓
Exam coherence	✓	✓	✓
Concentration coherence	✓	✓	✓
Required keywords	✓	✓	✓
2–3 sentences	✓	✓	✓
Cause-effect	✓	✓	✓
Positive tone	✓	✓	✓

Best Choice: Response 2

Since your exam is tomorrow, take a short break and then study in focused 25-minute sessions to improve your concentration. You can make good progress tonight, so stay confident and focus on the most important topics first.

Justification

Response 2 is the strongest because it provides a specific actionable study strategy rather than general advice.

The first sentence establishes a cause-effect relationship:

Break + Focused Study → Better Concentration

The second sentence provides encouragement and prioritizes important topics.

Therefore, it satisfies both the dialog act requirements and the constraint requirements.

2(d) Effect of Violating Two Constraints

Violation 1: Entity Coherence

Suppose the response says:

“Take a break and try to relax before your presentation.”

The conversation is about an exam, not a presentation.

This introduces an unrelated entity and breaks discourse coherence.

Effect:

- The response appears disconnected.
- The system seems not to understand the conversation.
- The advice becomes less useful.

Violation 2: Response Length

Suppose the system produces five or six sentences instead of 2–3.

Effect:

- It violates the explicit generation constraint.
- The response may become unnecessarily verbose.
- The user's immediate problem may become less clear.
- It may reduce conversational efficiency.

Therefore, constraint satisfaction is important for controlled dialogue generation.

3. Word Sense Disambiguation and Discourse Representation

Given Sentence

“The bank by the river flooded after the storm, but it was saved by quick action.”

The word bank is ambiguous.

It can mean:

1. A financial institution.
2. The land beside a river.

3(a) Resolve the Ambiguous Word “Bank”

The sentence contains:

“bank by the river”

This provides strong contextual information.

A financial institution can be located near a river, but the phrase “**by the river**” strongly suggests the geographical meaning.

More importantly:

“the bank ... flooded”

A riverbank can flood naturally.

A financial institution does not itself naturally “flood” as an entity in the same sense.

Therefore:

bank=riverbank

Constraint-Based Reasoning

Constraint 1: Context

“by the river”

Strongly favours riverbank.

Constraint 2: Semantic compatibility

flood(riverbank)

is highly natural.

Constraint 3: World knowledge

Rivers overflow and flood their banks.

Constraint 4: Coherence

The two clauses describe an environmental event and the subsequent rescue/protection of the affected bank.

Therefore:

Bank = riverbank

3(b) Predicate Logic Representation

Let:

- b = the riverbank
- r = river
- s = storm
- a = quick action

We can define:

Bank(b)
River(r)
Location (b, r)
Storm(s)
Flooded(b)

After (Flooded(b), Storm(s))

Quick Action(a)

Saved(b, a)

A combined representation is:

Bank(b) $\wedge$ River(r) $\wedge$ Location(b,r)

Flooded(b) $\wedge$ After(Flooded(b),Storm(s))

Quick Action(a) $\wedge$ Saved(b,a)

More Complete Logical Representation

The complete meaning can be represented as:

$\exists b,r,s,a[\text{Bank}(b) \wedge \text{River}(r) \wedge \text{Location}(b,r) \wedge \text{Storm}(s) \wedge \text{Flooded}(b) \wedge \text{After}(\text{Flooded}(b),s) \wedge \text{QuickAction}(a) \wedge \text{Saved}(b,a)]$

This captures the major entities and relations without changing the intended meaning.

3(c) Simple English Paraphrase

A clear simple-English paraphrase is:

The riverbank flooded after the storm, but quick action prevented further damage and saved the riverbank.

This resolves the ambiguity explicitly by replacing bank with riverbank.

3(d) RST-Style Discourse Tree

The sentence contains two major clauses:

Clause 1

The bank by the river flooded after the storm.

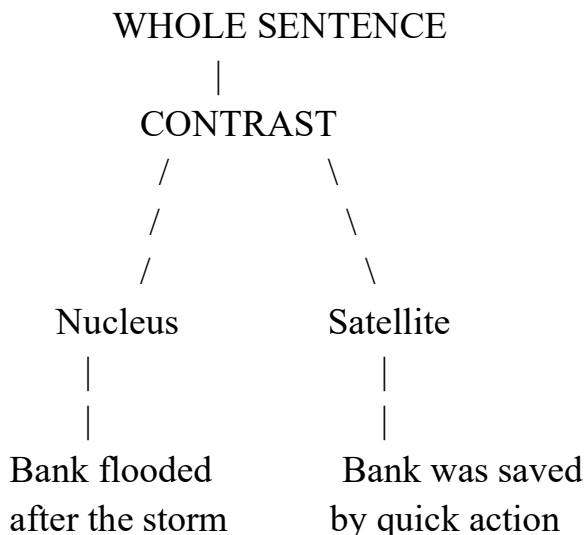
Clause 2

It was saved by quick action.

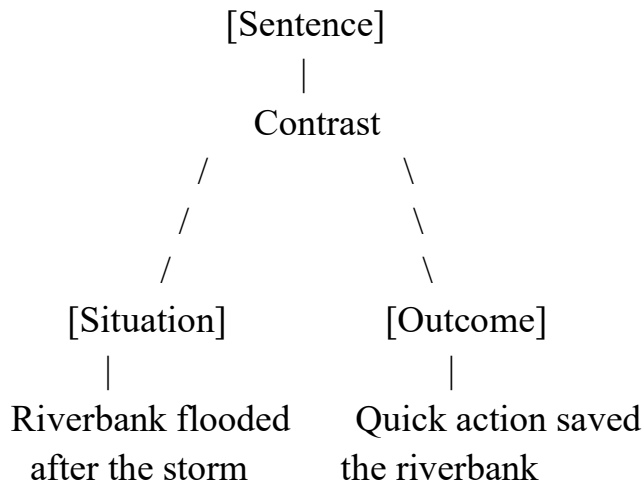
The connective:

But indicates a Contrast relation.

A simple RST representation is:



A more detailed representation:



The contrast relation can be interpreted as:

Although the riverbank was flooded, quick action prevented the situation from becoming worse / helped save it.

3(e) Constraint-Based Approach vs Pure Statistical Machine Translation

A constraint-based approach has several advantages for this sentence.

1. Better Word Sense Disambiguation

The word bank has multiple meanings.

A constraint-based system can explicitly consider:

bank_{financial} and bank_{river}

and eliminate the financial sense because:

flood(riverbank)

is semantically more compatible.

A purely statistical system may select a sense based on learned probabilities, which may not always preserve the intended meaning.